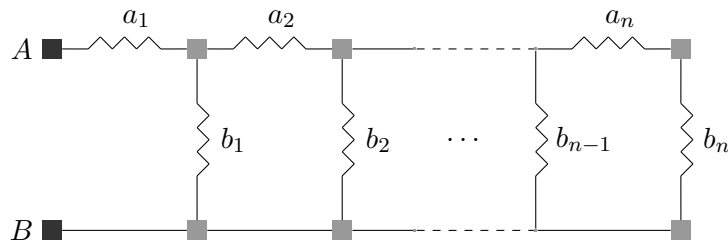


Lab Five: Optional Exercises

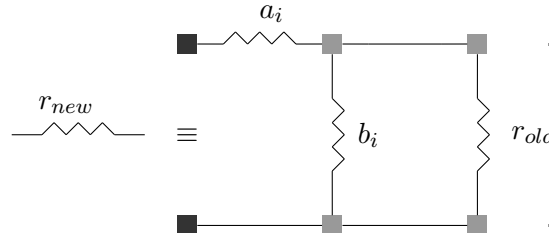
ID1063: Introduction to Programming

1. A **step-ladder** arrangement of resistors is shown below.



Write a program which accepts the values of the resistances (a_i s and b_i s as in the picture) in two arrays, and prints the effective resistance between A and B . Assume that all values are in some common unit, eg., ohms.

Hint: Update the value of the effective resistance from right-to-left as shown below.



When two resistances of values r_1, r_2 are combined in parallel, the effective resistance is $\frac{r_1 r_2}{r_1 + r_2}$, and when they are in series, the effective resistance is $r_1 + r_2$.

Example runs:

Enter the value of n: 2

Enter the value of the a_i : 2 2

Enter the value of the b_i s: 1 1

The effective resistance is: 2.75.

Enter the value of n: 3

Enter the value of the a_i : 1 2 3

Enter the value of the b_i s: 1 1 1

The effective resistance is: 1.7368.